SHIELD-MATLAB

Output Screenshots

Self-Healing Industrial Equipment and Link Diagnostics Using MATLAB Simulation

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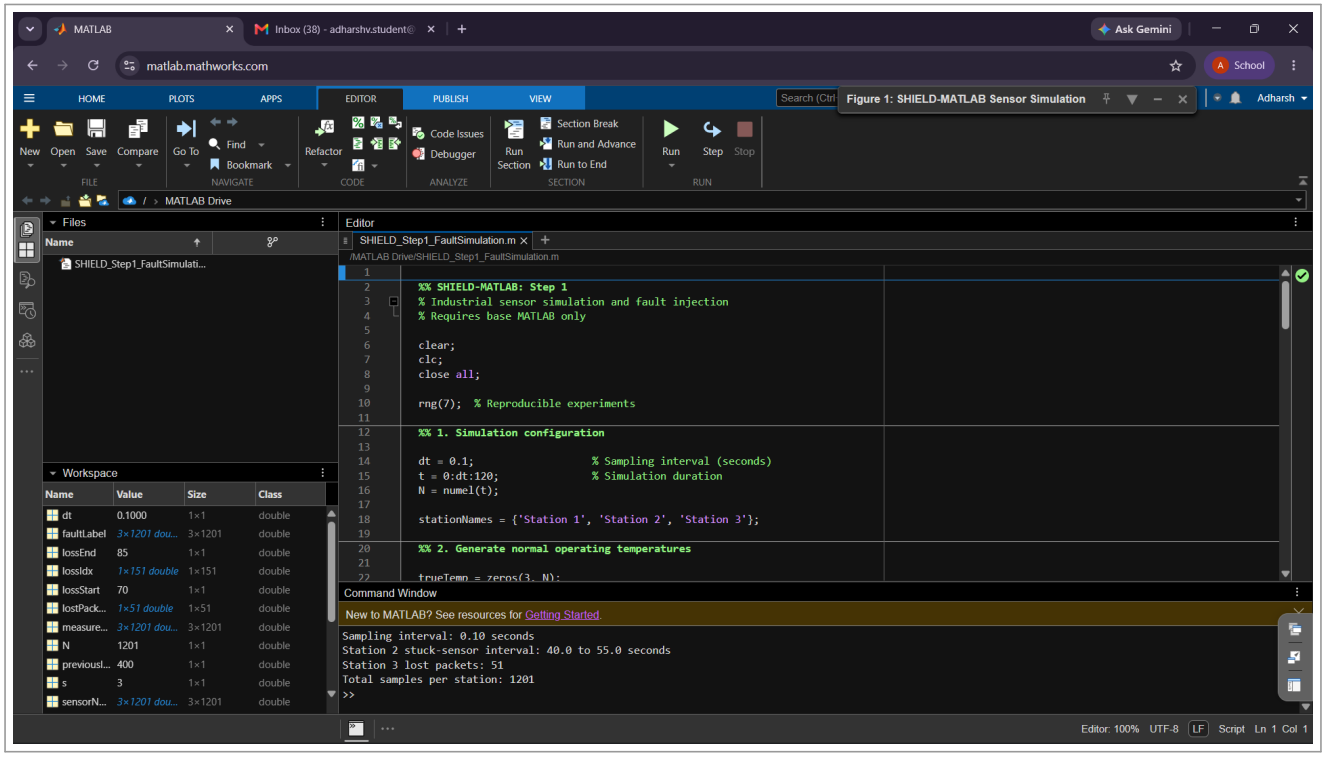
Software: MATLAB Online (base MATLAB)

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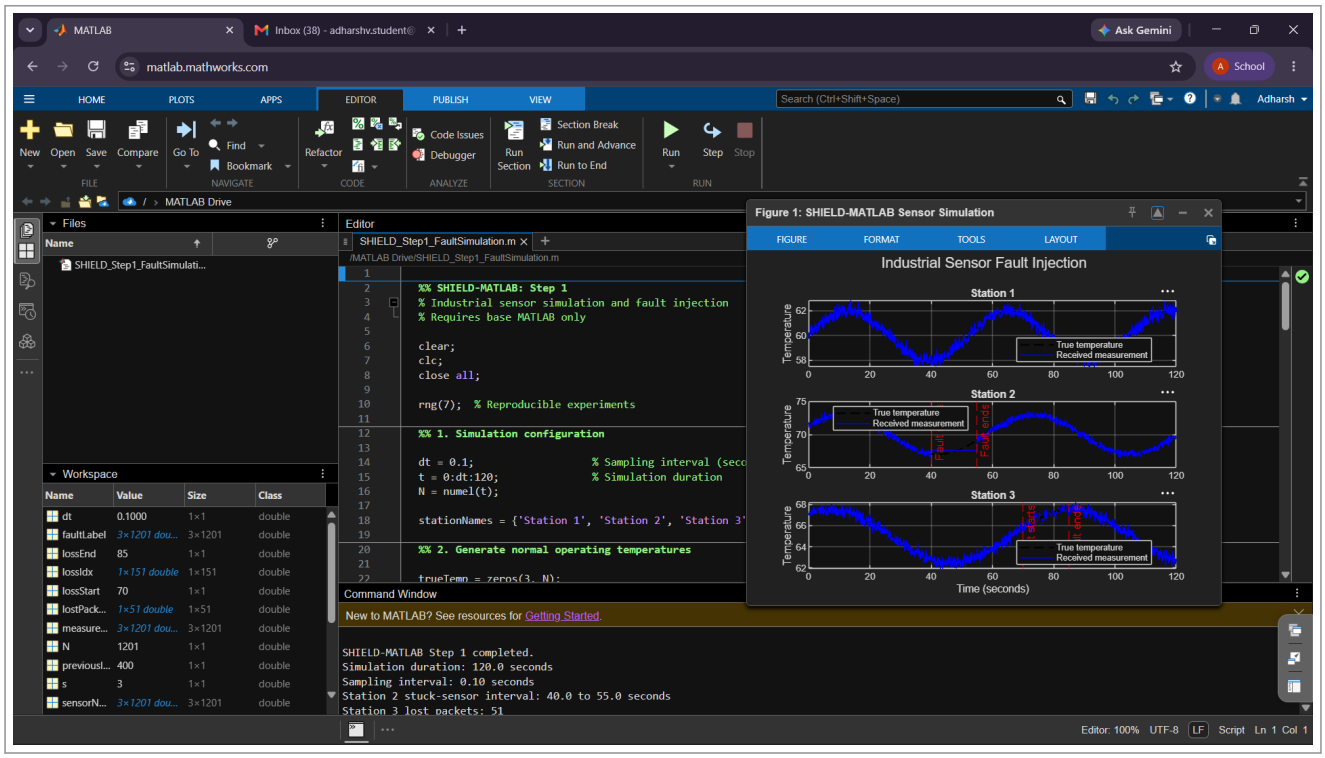
Step 1 - Sensor Simulation and Fault Injection

1.1 Source Code and Console Output



What this shows: Script setup, sampling interval 0.10 s, Station 2 stuck-sensor interval 40-55 s, Station 3 lost packets = 51, 1201 samples per station.

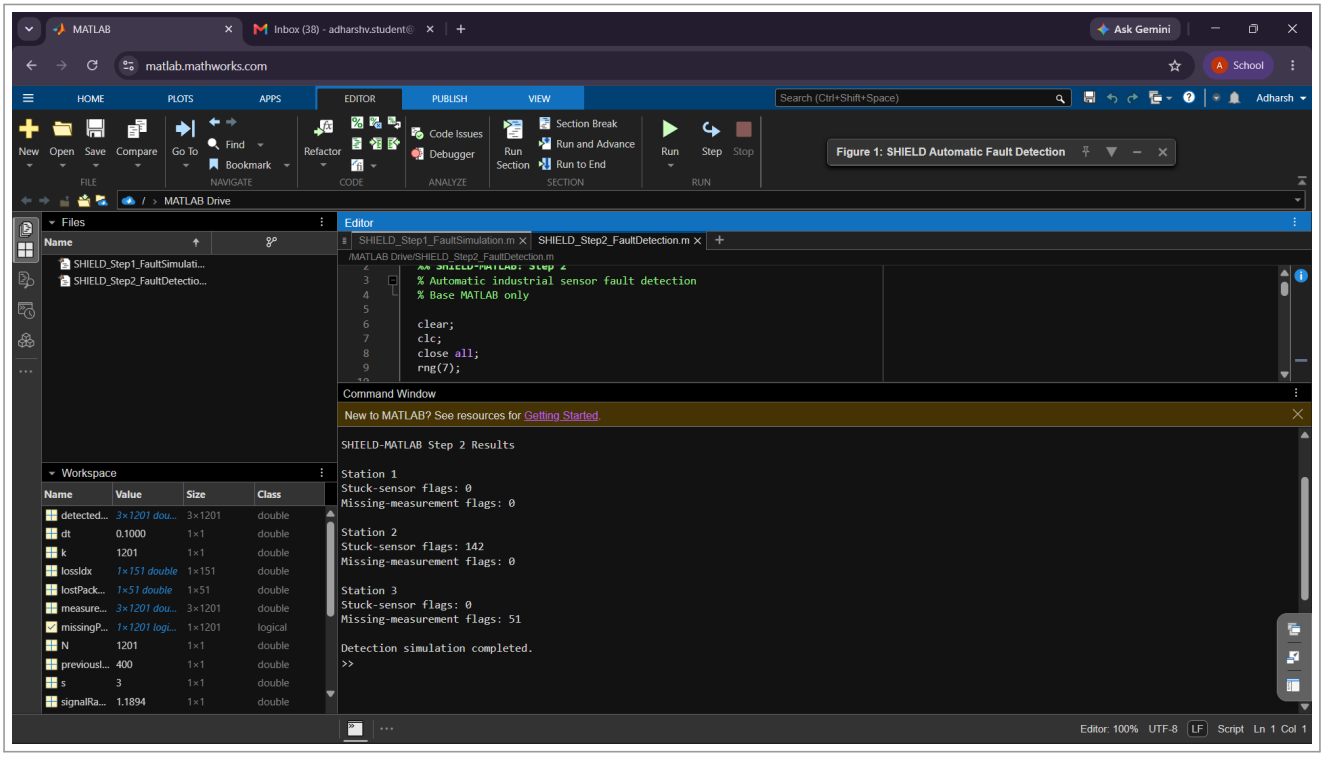
1.2 Industrial Sensor Fault Injection Figure



What this shows: True temperature (black dashed) versus received measurement (blue) for three stations. Red lines mark fault start and end; Station 2 shows the stuck sensor, Station 3 the missing data.

Step 2 - Automatic Sensor Fault Detection

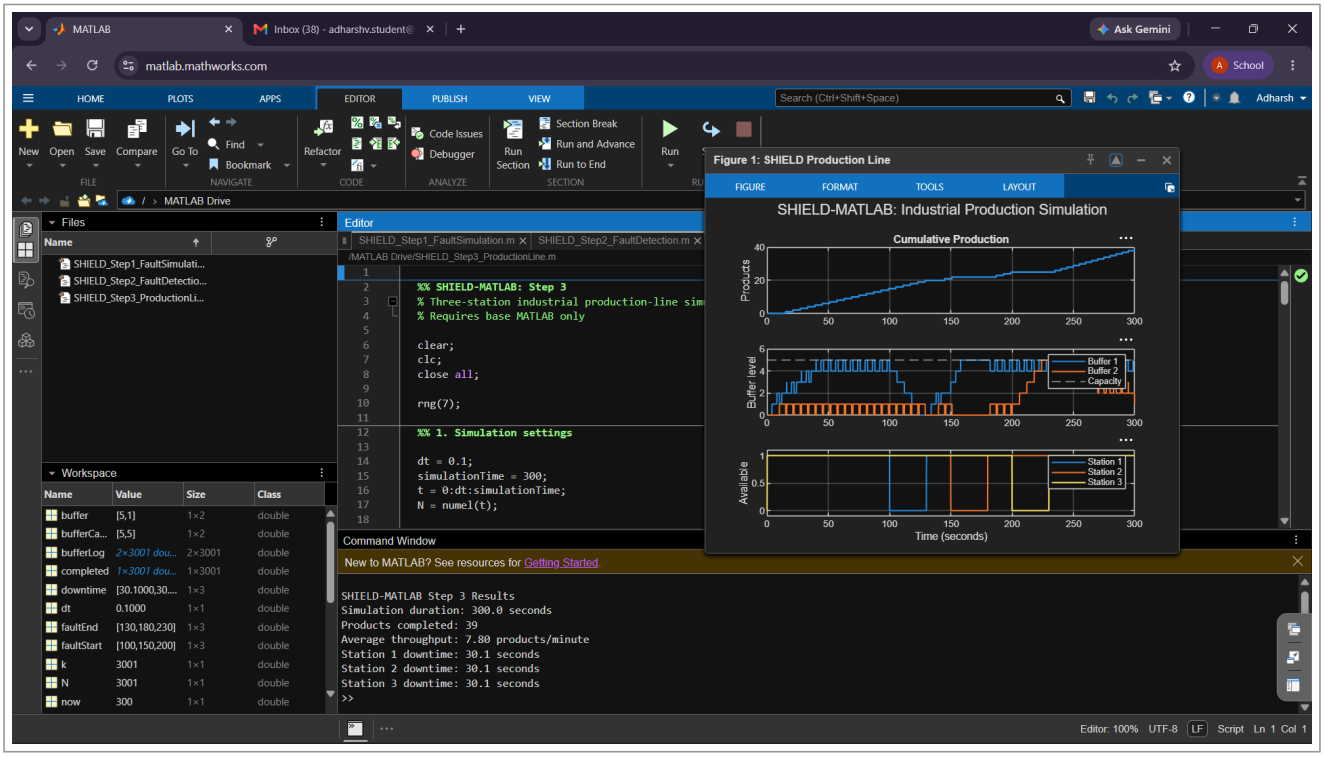
2.1 Detection Results



What this shows: Station 1: 0 / 0 flags. Station 2: 142 stuck-sensor flags. Station 3: 51 missing-measurement flags.

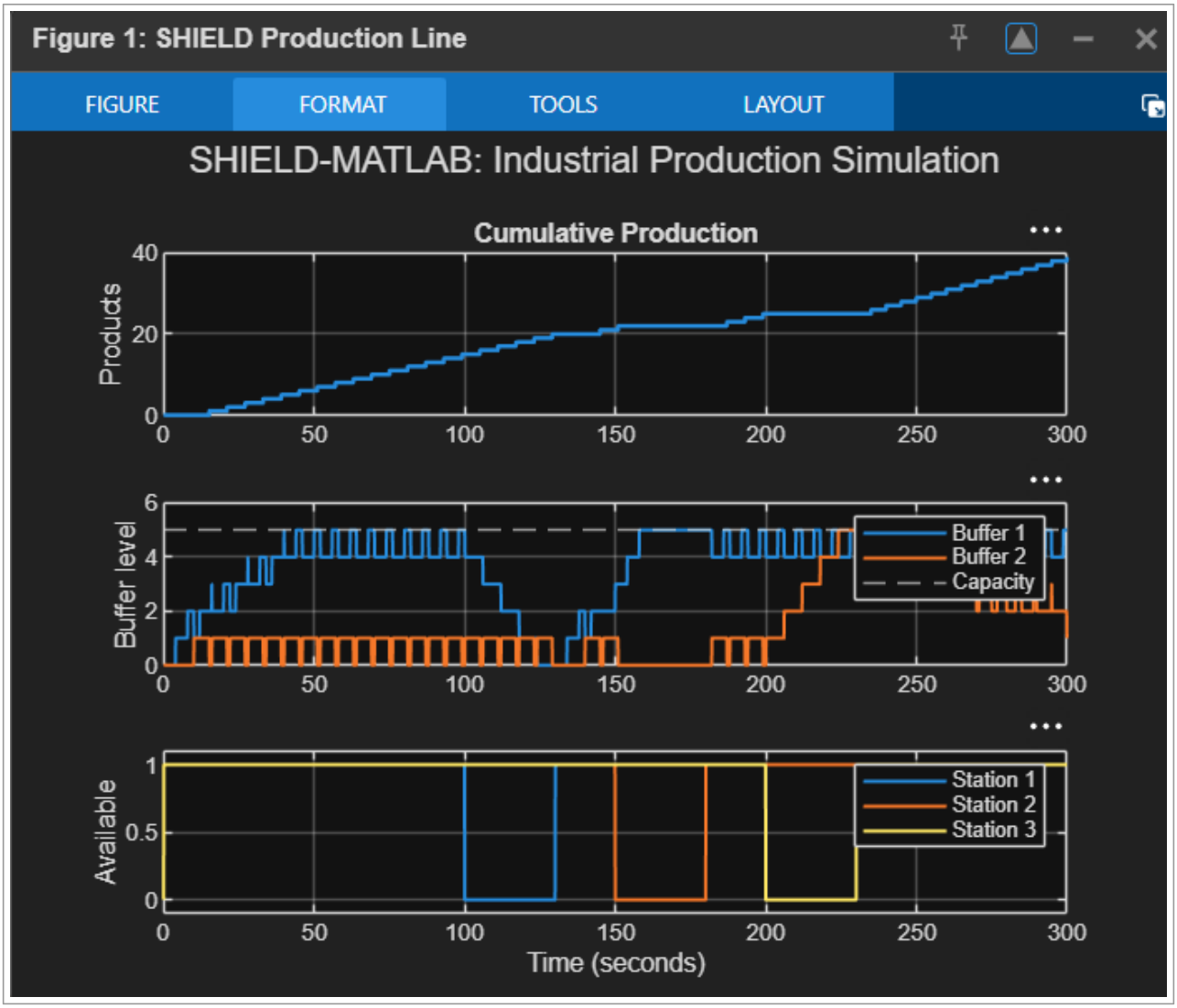
Step 3 - Three-Station Production Line

3.1 Console Results with Figure



What this shows: 300 s run: 39 products completed, 7.80 products/minute, 30.1 s downtime for each station.

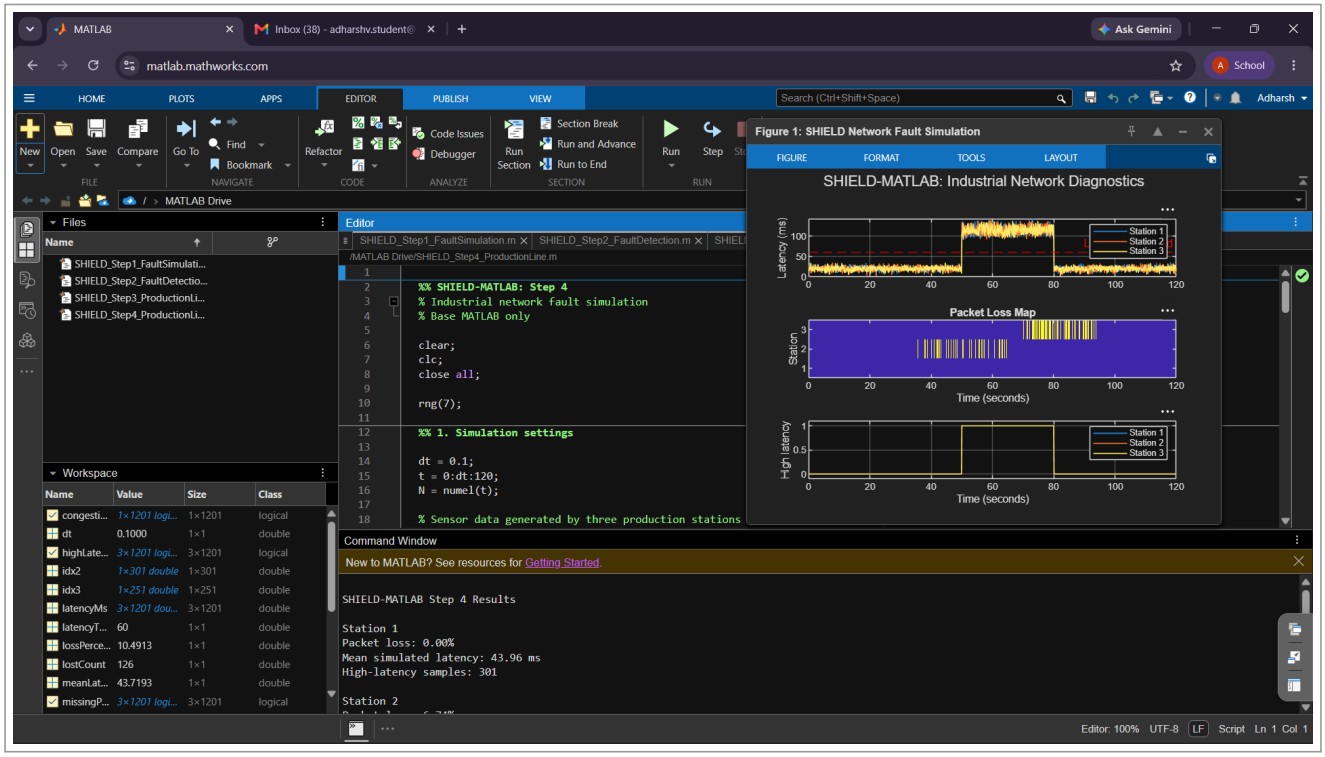
3.2 Production Line Figure



What this shows: Top: cumulative production. Middle: buffer levels against capacity 5. Bottom: station availability (1 = working, 0 = down).

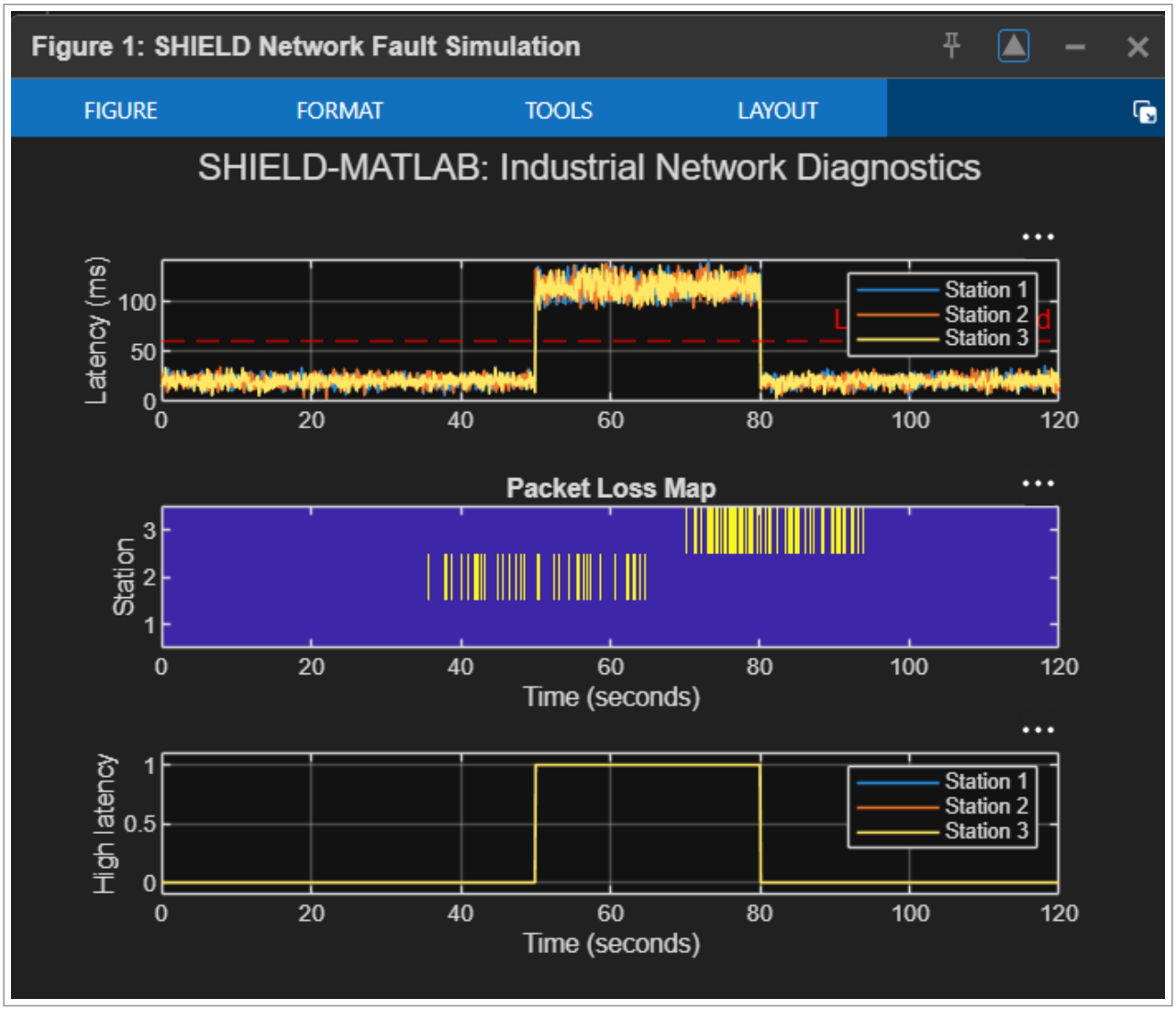
Step 4 - Network Fault Simulation

4.1 Network Diagnostics with Code and Workspace



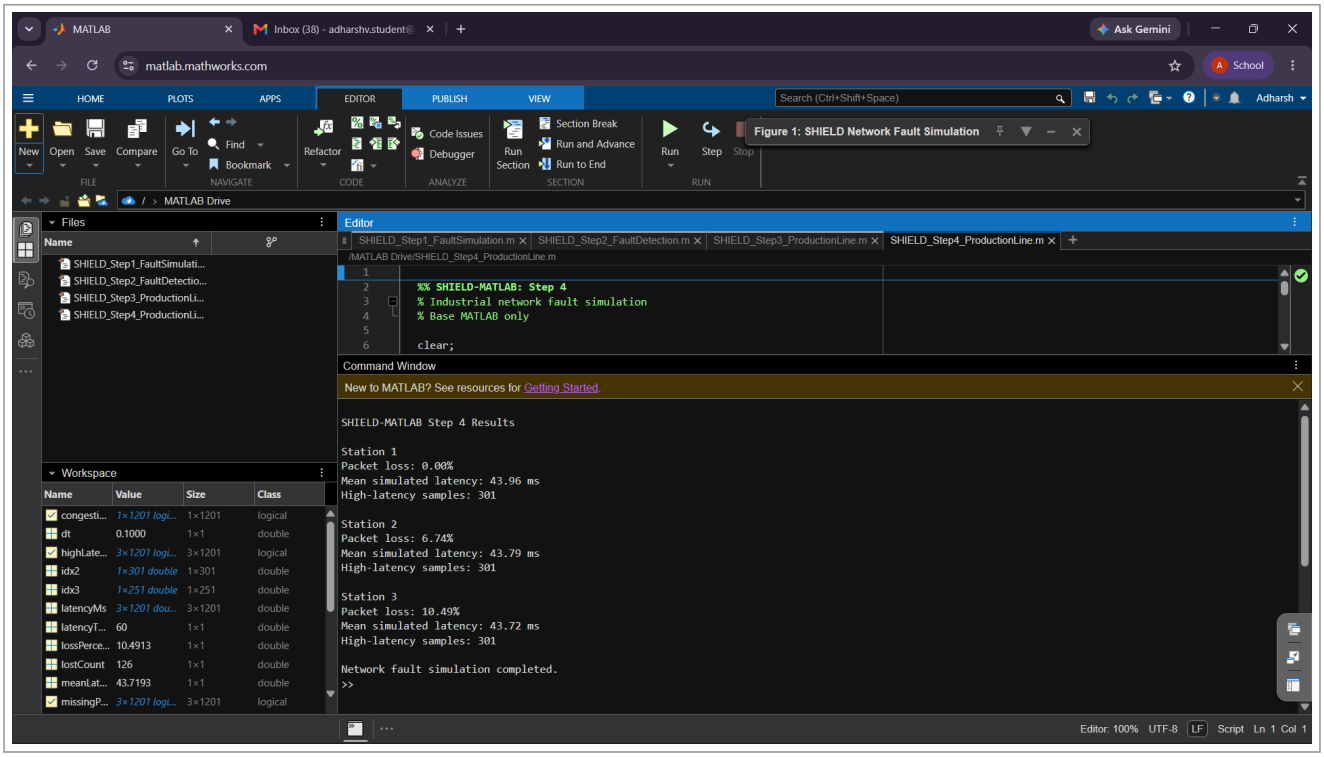
What this shows: Latency, packet-loss map and high-latency flag plotted alongside the Step 4 script and workspace variables.

4.2 Network Diagnostics Figure



What this shows: Latency rises above the 100 ms level between 50 s and 80 s; yellow lines in the packet-loss map are lost packets at Stations 2 and 3.

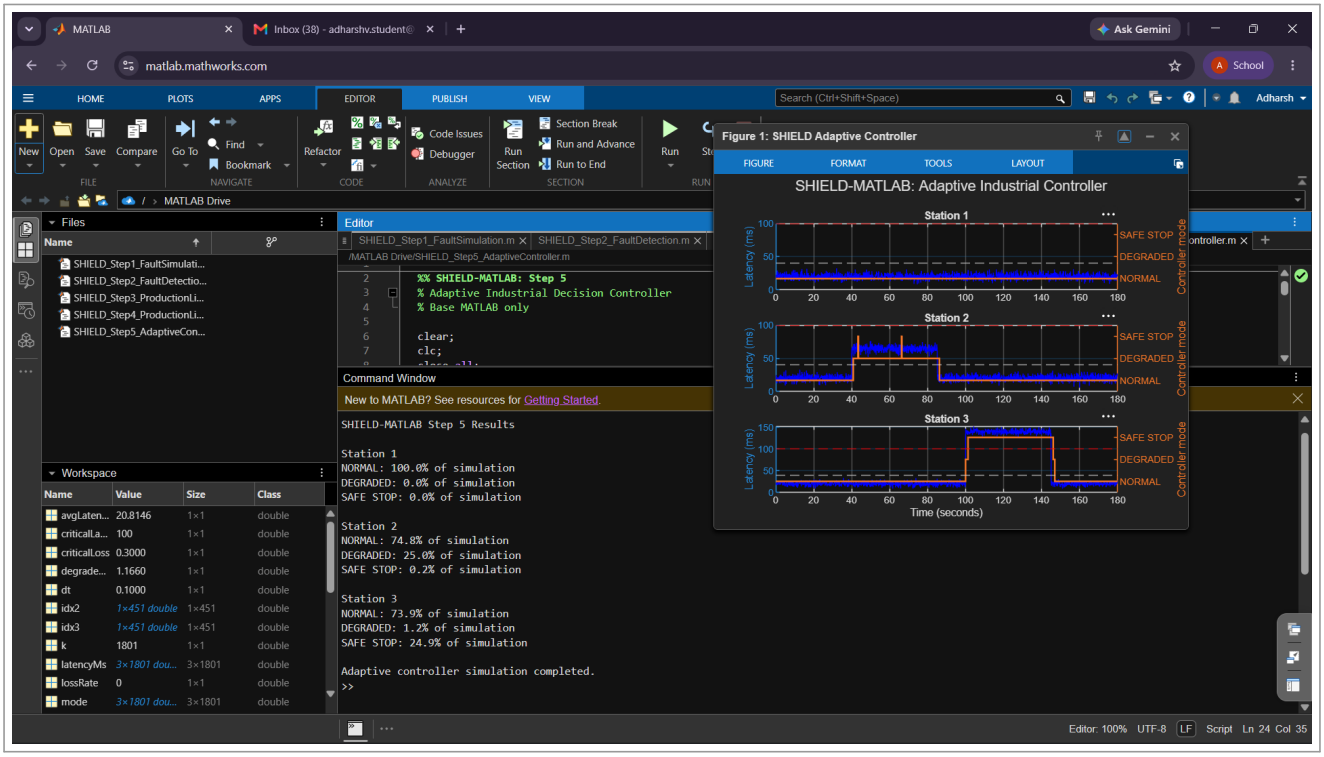
4.3 Console Results



What this shows: Packet loss 0.00 % / 6.74 % / 10.49 %; mean latency about 44 ms; 301 high-latency samples for each station.

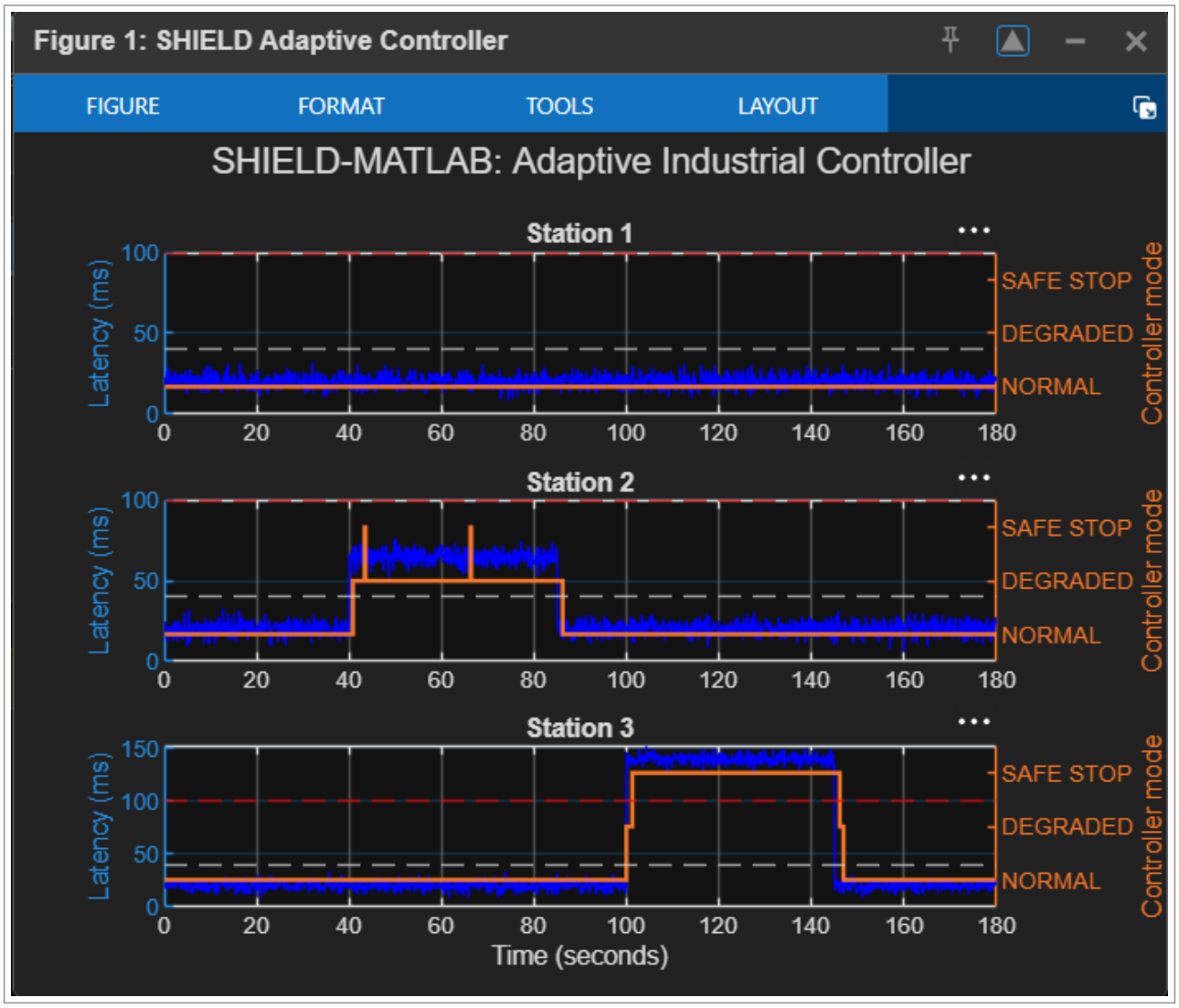
Step 5 - Adaptive Controller

5.1 Console Results with Figure



What this shows: Share of time in each mode. Station 2: NORMAL 74.8 %, DEGRADED 25.0 %, SAFE STOP 0.2 %. Station 3: NORMAL 73.9 %, DEGRADED 1.2 %, SAFE STOP 24.9 %.

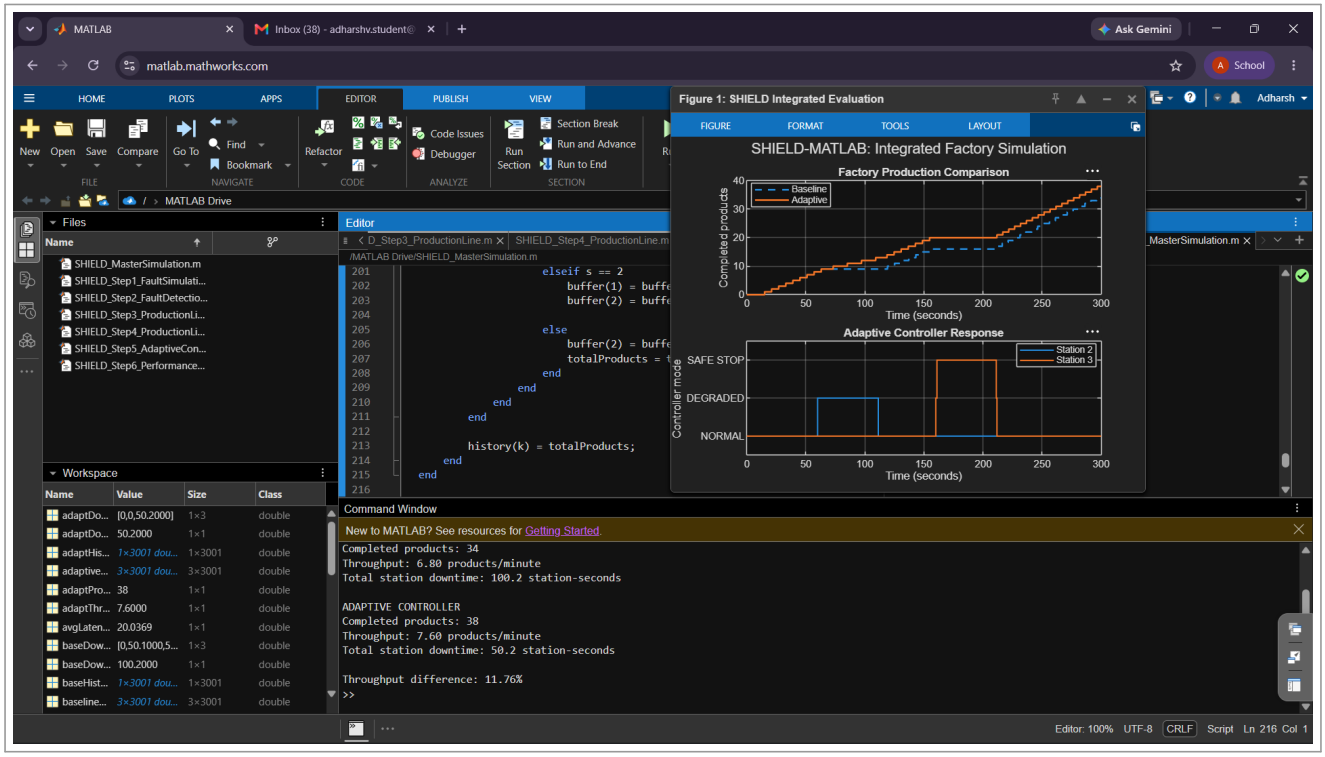
5.2 Adaptive Controller Figure



What this shows: Blue = latency, orange = controller mode, dashed lines = 40 ms reference and 100 ms critical threshold.

Step 6 - Integrated Baseline vs Adaptive Evaluation

6.1 Final Comparison Results and Figure



What this shows: Baseline: 34 products, 6.80/min, 100.2 station-s downtime. Adaptive: 38 products, 7.60/min, 50.2 station-s. Throughput difference 11.76 %.